

HCSM-03

Lorentzian Signature from the Hodge Star

The Hodge Star on $\ker D$ and the Canonical Lorentzian Metric on the Four Harmonic Forms

Stanley Preschutti

Entropy Research Institute / Information Physics Institute

ORCID: 0009-0004-5445-1744

scstanp@yahoo.com

September 2026

Abstract

This paper constructs a Lorentzian metric of signature $(-, +, +, +)$ on the four dimensional kernel $\ker D$ of the Hodge-Dirac operator on the periodic 8×8 torus.

The Hodge star restricted to $\ker D$ has eigenvalues $\{+1, -1, +i, -i\}$: the ± 1 pair comes from the $(H^0 \oplus H^2)$ sector, the $\pm i$ pair from the H^1 sector. We construct a symmetric bilinear form

$$g = Q|_{H^0 \oplus H^2} \oplus \langle \cdot, \cdot \rangle|_{H^1}$$

where $Q(\alpha, \beta) = \int \alpha \wedge \beta$ is the intersection form on the even degree sector and $\langle \cdot, \cdot \rangle$ is the Hodge inner product on the odd-degree sector. The metric g has signature $(3, 1)$ and is D_4 -invariant.

The two structural inputs are made explicit: (i) the use of Q on $H^0 \oplus H^2$ rather than the Hodge inner product (which would give a Euclidean signature), and (ii) the identification of the resulting metric with the emergent spacetime metric. The first is fixed by the additional condition that the pure degree forms v_0 and v_2 have zero norm under g ; the second is a framework level disclosure of the same type as the identification $d = \dim \ker D$. Given these, the metric is unique up to scale among D_4 -invariant symmetric forms on $\ker D$. The timelike direction is $(v_0 - v_2)/\sqrt{2}$.

The construction is the discrete analogue of the Hodge–Riemann bilinear relations on the cohomology of a Kähler manifold. All numerical claims are verified at machine precision.

Contents

1	Introduction	2
1.1	The signature question	2
1.2	Summary of results	2
1.3	Disclosed inputs	3
2	Setup	3
2.1	Substrate and Hodge complex	3
2.2	Kernel basis	3
3	The Hodge star on $\ker D$	3
3.1	The discrete Hodge star	3
3.2	The Hodge star spectrum on $\ker D$	4

4	The canonical Lorentzian metric	4
4.1	The two canonical bilinear forms	4
4.2	The metric construction	4
5	Conditional uniqueness	5
5.1	The space of D_4 -invariant symmetric forms	5
5.2	The two conditions	5
6	The timelike direction	6
7	D_4 invariance and decomposition	6
8	Consequences	6
8.1	Status of the derivation	6
8.2	No Wick rotation	7
8.3	Relation to the spin-statistics structure	7
9	Open items	7
10	Reproducibility	7
11	Conclusion	8

1 Introduction

1.1 The signature question

The Hodge Complex Standard Model derives the spacetime dimension $d = 4$ from the Hodge–Dirac kernel on the periodic 8×8 torus:

$$\dim \ker D = b_0 + b_1 + b_2 = 1 + 2 + 1 = 4. \quad (1)$$

This is established in HCSM-02 (Theorem 6.1.1). The four harmonic forms of $\ker D$ are the four emergent spacetime directions (by the framework’s disclosed input S3 of HCSM-01).

What remains is the signature of the metric on $\ker D$. The Hodge complex of differential forms on Λ is equipped with a positive-definite Euclidean structure, so the natural metric on $\ker D$ inherited from the Hodge inner product is positive-definite. A Lorentzian metric requires a structural choice: at least one of the four directions must be distinguished.

This paper constructs a specific D_4 -invariant Lorentzian metric on $\ker D$ from the canonical bilinear forms of the Hodge complex, and shows that this metric is the unique one (up to scale) satisfying two structural conditions. The construction is the discrete realization of the Hodge–Riemann bilinear relations.

1.2 Summary of results

1. **Hodge star spectrum.** The Hodge star restricted to $\ker D$ has eigenvalues $\{+1, -1, +i, -i\}$, with ± 1 from $H^0 \oplus H^2$ and $\pm i$ from H^1 (§3).
2. **Canonical Lorentzian metric.** The metric $g = Q|_{H^0 \oplus H^2} \oplus \langle \cdot, \cdot \rangle|_{H^1}$ on $\ker D$ has signature $(3, 1)$ (§4).
3. **Conditional uniqueness.** Among all D_4 -invariant symmetric forms on $\ker D$, the metric g is the unique one (up to scale) that vanishes on the pure-degree forms v_0, v_2 and has signature $(3, 1)$ (§5).

4. **Timelike direction.** For the signature convention $(-, +, +, +)$, the timelike direction is $(v_0 - v_2)/\sqrt{2}$ (§6).
5. **D_4 invariance and decomposition.** The metric g is D_4 -invariant, and $\ker D \cong 2A_1 \oplus E$ as a D_4 representation (§7).

1.3 Disclosed inputs

Following HCSM-01 §1.2 and HCSM-02 §1.3, the following items are disclosed and carried through the paper:

- (S1) The substrate is $\Lambda = \mathbb{Z}_8 \times \mathbb{Z}_8$ with D_4 symmetry.
- (S2) The scalar Laplacian is the 5-point Laplacian.
- (S3) The emergent spacetime dimension is $\dim \ker D$.
- (S4) **Metric construction.** On $\ker D$, the even-degree sector $H^0 \oplus H^2$ is paired by the intersection form $Q(\alpha, \beta) = \int \alpha \wedge \beta$; the odd-degree sector H^1 is paired by the Hodge inner product. The mixed-degree pairings vanish.
- (S5) **Physical identification.** The bilinear form g is identified with the emergent spacetime metric on the four harmonic directions.

(S4) is fixed in §5 by the additional condition that g vanishes on the pure-degree forms. (S5) is a framework-level identification analogous to (S3): the metric constructed on $\ker D$ is assumed to be the emergent spacetime metric.

2 Setup

2.1 Substrate and Hodge complex

$\Lambda = \mathbb{Z}_8 \times \mathbb{Z}_8$, $N = 64$ sites. The Hodge complex has three graded pieces $\Omega^0, \Omega^1, \Omega^2$ of dimensions 64, 128, 64; total 256. The Hodge–Dirac operator $D = d + d^*$ satisfies $D^2|_{\Omega^k} = \Delta \cdot \text{id}_{\Omega^k}$ (HCSM-02 Theorem 3.2.1).

2.2 Kernel basis

By HCSM-02 Theorem 6.1.1, $\dim \ker D = 4$. We use the form-degree-aligned basis

$$\{v_0, v_2, v_1^A, v_1^B\}, \quad (2)$$

where v_0 is the constant 0-form (spanning H^0), v_2 is the constant 2-form (spanning H^2), and v_1^A, v_1^B are the two harmonic 1-forms (spanning H^1).

3 The Hodge star on $\ker D$

3.1 The discrete Hodge star

The Hodge star on Λ is

$$\star_0 = \text{id}_{\Omega^0}, \quad (\star_1 \omega)_x = \omega_y, \quad (\star_1 \omega)_y = -\omega_x, \quad \star_2 = \text{id}_{\Omega^2}. \quad (3)$$

It satisfies $\star_0^2 = +1$, $\star_1^2 = -1$, $\star_2^2 = +1$, and $[\star, D^2] = 0$.

Proposition 3.1 ($\star$ preserves the kernel). $\star(\ker D) = \ker D$.

Proof. Let $v \in \ker D$. Then $\Delta v = 0$. Since $[\star, \Delta] = 0$, $\Delta(\star v) = \star \Delta v = 0$, so $\star v \in \ker \Delta = \ker D$. $\square$

3.2 The Hodge star spectrum on $\ker D$

Theorem 3.2. *In the basis (2), the Hodge star restricted to $\ker D$ has matrix*

$$\star|_{\ker D} = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & -1 \\ 0 & 0 & 1 & 0 \end{pmatrix}. \quad (4)$$

Its eigenvalues are $\{+1, -1, +i, -i\}$.

Proof. The star exchanges H^0 and H^2 via $\star_0$ and $\star_2$, and rotates H^1 via $\star_1$. On the aligned basis:

- $\star v_0 = v_2, \star v_2 = v_0$: the block $\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ with eigenvalues ± 1 .
- $\star v_1^A = -v_1^B, \star v_1^B = v_1^A$: the block $\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$ with eigenvalues $\pm i$.

□

Remark 3.3. The complex structure on H^1 is $\star_1$ itself: $\star_1^2 = -1$ on H^1 , so $(\Omega^1|_{\lambda=4}, \langle \cdot, \cdot \rangle, \star_1)$ is a Hermitian vector space. This is the discrete analogue of the complex structure induced by the Hodge star on odd-degree cohomology.

4 The canonical Lorentzian metric

4.1 The two canonical bilinear forms

The Hodge complex carries two canonical bilinear forms:

- The **intersection form** $Q(\alpha, \beta) = \int_{\Lambda} \alpha \wedge \beta$, defined for α, β of complementary degrees. On the even-degree cohomology $H^0 \oplus H^2$, it pairs H^0 with H^2 .
- The **Hodge inner product** $\langle \alpha, \beta \rangle = \int_{\Lambda} \alpha \wedge \star \beta$, defined for α, β of the same degree.

4.2 The metric construction

On $\ker D = H^0 \oplus H^1 \oplus H^2$, define g blockwise:

$$g(\alpha, \beta) = \begin{cases} Q(\alpha, \beta), & \alpha, \beta \in H^0 \oplus H^2, \\ \langle \alpha, \beta \rangle, & \alpha, \beta \in H^1, \\ 0, & \text{mixed degrees.} \end{cases} \quad (5)$$

The block-diagonal structure is forced: Q vanishes on pairs of same-degree forms, and $\langle \cdot, \cdot \rangle$ vanishes on pairs of different-degree forms.

Theorem 4.1 (Lorentzian signature). *In the basis (2), the metric g has matrix*

$$g = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}. \quad (6)$$

Its eigenvalues are $\{-1, +1, +1, +1\}$: signature $(3, 1)$.

Proof. On $H^0 \oplus H^2$, $Q(v_0, v_2) = \int v_0 \wedge v_2 = 1$ (normalized volume), and $Q(v_0, v_0) = Q(v_2, v_2) = 0$ by degree. The block $\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ has eigenvalues ± 1 . On H^1 , the Hodge inner product in the orthonormal basis $\{v_1^A, v_1^B\}$ is the identity, giving eigenvalues $+1, +1$. Total signature: $(3, 1)$. □

5 Conditional uniqueness

The metric g is not the unique D_4 -invariant symmetric form on $\ker D$. We first characterize the full space of such forms, then state the two additional conditions that single out g .

5.1 The space of D_4 -invariant symmetric forms

Proposition 5.1. *The space of D_4 -invariant symmetric bilinear forms on $\ker D$ is 4-dimensional. In the basis (2), a general element is*

$$G = \begin{pmatrix} a & b & 0 & 0 \\ b & c & 0 & 0 \\ 0 & 0 & d & 0 \\ 0 & 0 & 0 & d \end{pmatrix}, \quad (a, b, c, d) \in \mathbb{R}^4. \quad (7)$$

The four parameters are the coefficients on the basis elements $\{E_0 = \text{diag}(1, 0, 0, 0), E_1 = \frac{1}{\sqrt{2}}(\delta_{v_0 v_2}), E_2 = \text{diag}(0, 1, 0, 0), E_3 = I_{E\text{-block}}\}$.

Proof. Apply Schur's lemma to the D_4 representation $\ker D = 2A_1 \oplus E$. The two copies of A_1 span $\{v_0, v_2\}$, on which a D_4 -invariant bilinear form is a general 2×2 symmetric matrix (4 real parameters minus the antisymmetric part, which vanishes). The E block is irreducible, so its invariant form is a scalar multiple of the identity; in 2 dimensions, the antisymmetric part vanishes for a symmetric form, leaving one scalar d . Total: 3 parameters from the A_1 block plus 1 from $E = 4$. $\square$

The HCSM-03 metric g of Theorem 4.1 corresponds to $(a, b, c, d) = (0, 1, 0, 1)$.

5.2 The two conditions

Two structural conditions single out $(a, b, c, d) = (0, 1, 0, 1)$ up to scale:

(C1) Vanishing on pure-degree forms: $g(v_0, v_0) = g(v_2, v_2) = 0$. This is the discrete analogue of the Hodge–Riemann bilinear relations, in which primitive forms of pure Hodge type (p, p) pair to zero.

(C2) Lorentzian signature: the metric has signature $(3, 1)$ or $(1, 3)$.

Theorem 5.2 (Conditional uniqueness). *Among all D_4 -invariant symmetric forms G on $\ker D$, the HCSM-03 metric g is the unique one (up to overall scale) that satisfies both conditions (C1) and (C2).*

Proof. Condition (C1) forces $a = c = 0$, leaving $G = \begin{pmatrix} 0 & b & 0 \\ b & 0 & 0 \\ 0 & 0 & dI_2 \end{pmatrix}$. The even-degree block has eigenvalues $\pm b$, and the odd-degree block contributes d twice. Condition (C2) requires $b \neq 0$ and $d \neq 0$ with $\text{sign } b \cdot \text{sign } d > 0$ (so that the odd-degree pair has the same sign as the positive eigenvalue of the even-degree block). Up to overall scale, the metric is fixed: $d/b > 0$ is a scale choice, and normalization $d = b = 1$ gives the HCSM-03 metric. $\square$

Remark 5.3. The parameters (a, b, c, d) in (7) are not free continuous parameters of a physical theory. They are the coefficients of a symmetric bilinear form on a 4-dimensional vector space. The two conditions (C1), (C2) reduce the 4-parameter family to a single point up to scale. The metric g is therefore "canonical" in the honest sense: among all D_4 -invariant forms on $\ker D$ satisfying the two structural conditions, it is unique.

Remark 5.4 (Hodge–Riemann analogy). On a compact Kähler manifold, the Hodge–Riemann bilinear relations state that the natural bilinear form on the middle-degree cohomology has signature (b^+, b^-) determined by the Hodge numbers, with the primitive forms of pure Hodge type (p, p) pairing to zero. On the 2-torus T^2 , the cohomology decomposes as $H^0 \oplus H^1 \oplus H^2$ with $H^0 \oplus H^2$ of even degree, and the intersection form on $H^0 \oplus H^2$ is the discrete analogue of the Hodge–Riemann form on middle-degree cohomology. The HCSM-03 metric is the direct discrete realization of this classical structure.

6 The timelike direction

Corollary 6.1. *With the signature convention $(-, +, +, +)$ of Theorem 4.1 (eigenvalues $\{-1, +1, +1, +1\}$), the timelike direction is*

$$\mathbf{t} = \frac{v_0 - v_2}{\sqrt{2}}. \quad (8)$$

The three spacelike directions are $(v_0 + v_2)/\sqrt{2}$, v_1^A , v_1^B .

Proof. Direct computation: $g(v_0 - v_2, v_0 - v_2)/2 = (Q(v_0, v_0) - 2Q(v_0, v_2) + Q(v_2, v_2))/2 = (0 - 2 + 0)/2 = -1$. $g(v_0 + v_2, v_0 + v_2)/2 = +1$. $g(v_1^A, v_1^A) = g(v_1^B, v_1^B) = +1$. Total: one timelike, three spacelike. $\square$

Remark 6.2. The sign convention is fixed by the choice of which eigenvalue carries the negative sign. With the opposite convention $(+, -, -, -)$, the roles are reversed and the timelike direction is $(v_0 + v_2)/\sqrt{2}$. Both are representations of the same metric; only the labelling of timelike and spacelike is swapped. Throughout the paper, we adopt $(-, +, +, +)$.

7 D_4 invariance and decomposition

Proposition 7.1. *As a D_4 representation, $\ker D \cong 2A_1 \oplus E$. In the basis (2), the generators r and s act by*

$$R_{\ker} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & -1 & 0 \end{pmatrix}, \quad S_{\ker} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix}. \quad (9)$$

Proof. The subspace $\text{span}\{v_0, v_2\}$ is fixed pointwise by both generators, giving two copies of the trivial representation A_1 . The subspace $\text{span}\{v_1^A, v_1^B\}$ carries the 2-dimensional irreducible representation E . $\square$

Theorem 7.2. *The metric g is invariant under the D_4 action on $\ker D$: $R_{\ker}^T g R_{\ker} = g$ and $S_{\ker}^T g S_{\ker} = g$.*

Proof. Direct computation. The verification at machine precision gives $\|R_{\ker}^T g R_{\ker} - g\|_F \sim 7 \times 10^{-16}$ and $\|S_{\ker}^T g S_{\ker} - g\|_F \sim 6 \times 10^{-16}$. $\square$

8 Consequences

8.1 Status of the derivation

Before this paper, the framework had:

- $d = 4$ derived from $\dim \ker D = 4$ (HCSM-01, HCSM-02).
- Lorentzian signature $(3, 1)$ asserted by analogy with the photon-helicity anchor (HCSM-01).

After this paper:

- $d = 4$ derived from $\dim \ker D = 4$.
- Lorentzian signature $(3, 1)$ derived from the Hodge star on $\ker D$ and the two structural conditions (C1), (C2), given the disclosed metric construction (S4) and physical identification (S5).

The photon-helicity anchor is now a consistency check rather than the primary derivation of the signature.

8.2 No Wick rotation

The metric g is constructed directly from the intersection form and the Hodge inner product on the harmonic forms. No Wick rotation is used: the Lorentzian signature emerges from the indefinite signature of the intersection form on $H^0 \oplus H^2$, not from an analytic continuation of a Euclidean metric. This is a technical statement about the construction, not a physical claim.

8.3 Relation to the spin-statistics structure

The Hodge complex carries a canonical $\mathbb{Z}_2$ grading: even forms $\Omega^0 \oplus \Omega^2$ (bosonic) versus odd forms Ω^1 (fermionic), per HCSM-02 §4. On $\ker D = 2A_1 \oplus E$, the timelike direction $(v_0 - v_2)/\sqrt{2}$ lies entirely in the even sector. The three spacelike directions include the even combination $(v_0 + v_2)/\sqrt{2}$ and the two odd-degree harmonic 1-forms. The extension of this structure to Standard Model spin and chirality is the subject of HCSM-42.

9 Open items

- The physical identification (S5) — that g is the emergent spacetime metric — is a framework-level disclosure. Its justification would require the continuum limit of the framework (item 6.14).
- The connection of the HCSM-03 construction to the classical Hodge–Riemann bilinear relations is observed (§5, Remark 5.4) but not developed. A discrete proof that the HCSM-03 form coincides with the classical Hodge–Riemann form under the discrete-to-continuum scaling is a program item.

10 Reproducibility

All numerical claims in this paper are verified at machine precision in IEEE-754 double precision. The companion script `HCSM-03_verification.py` constructs the discrete Hodge complex on $\Lambda = \mathbb{Z}_8 \times \mathbb{Z}_8$, computes the Hodge star and Hodge–Dirac operators, restricts the star to $\ker D$, and derives the metric g . It verifies:

1. Nilpotency $d_1 d_0 = 0$.
2. Kernel dimension $\dim \ker D = 4$.
3. Hodge star involution $\star^2 = \text{diag}(+1, -1, +1)$.
4. Commutation $[\star, D^2] = 0$.
5. Star spectrum $\text{spec}(\star|_{\ker D}) = \{+1, -1, +i, -i\}$.
6. Metric g with signature $(3, 1)$.
7. Timelike direction $(v_0 - v_2)/\sqrt{2}$.

8. D_4 invariance of g .
9. The dimension of the D_4 -invariant symmetric-form space on $\ker D$ is 4.
10. The HCSM-03 metric is the unique element of this 4-dimensional space satisfying conditions (C1), (C2) up to scale.

Running `12-VERIFICATION/run_all.py` reproduces every numerical claim in the HCSM paper stack.

11 Conclusion

We have constructed a Lorentzian metric of signature $(3, 1)$ on the 4-dimensional kernel $\ker D$ of the Hodge–Dirac operator on the periodic 8×8 torus. The metric is $g = Q|_{H^0 \oplus H^2} \oplus \langle \cdot, \cdot \rangle|_{H^1}$, where Q is the intersection form on the even-degree sector and $\langle \cdot, \cdot \rangle$ is the Hodge inner product on the odd-degree sector.

The Hodge star on $\ker D$ has eigenvalues $\{+1, -1, +i, -i\}$, splitting into a real involution on $H^0 \oplus H^2$ and a complex structure on H^1 . The metric g is D_4 -invariant and is the unique element of the 4-dimensional space of D_4 -invariant symmetric forms on $\ker D$ satisfying the two structural conditions (C1) $g(v_0, v_0) = g(v_2, v_2) = 0$ and (C2) signature $(3, 1)$. The construction is the discrete realization of the classical Hodge–Riemann bilinear relations on cohomology.

Given the metric construction (S4) and the physical identification (S5), the framework now derives both $d = 4$ and the Lorentzian signature $(3, 1)$ from the substrate’s Hodge structure.

References

- [1] S. Preschutti, *HCSM-00: Foundations, Notation, and Mapping*, HCSM White Paper Series (2026).
- [2] S. Preschutti, *HCSM-01: The 8×8 Torus Substrate from $d = 4$* , HCSM White Paper Series (2026).
- [3] S. Preschutti, *HCSM-02: The Hodge Complex on Λ* , HCSM White Paper Series (2026).
- [4] S. Preschutti, *HCSM-04: The 14-Mode Multiplet at $\lambda = 4$* , HCSM White Paper Series (2026).
- [5] S. Preschutti, *HCSM-42: Spin Chirality $\mathbb{Z}_2$ Grading*, HCSM White Paper Series (2026).
- [6] J. Dodziuk, *Finite-difference approach to the Hodge theory of harmonic forms*, Amer. J. Math. **98**, 79 (1976).
- [7] M. Nakahara, *Geometry, Topology and Physics*, 2nd ed., CRC Press (2003).
- [8] C. Voisin, *Hodge Theory and Complex Algebraic Geometry I*, Cambridge University Press (2002). [Hodge–Riemann bilinear relations.]